Free-body diagrams

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Draw only the external forces on one chosen object.

SUCCESS CRITERIA - I can...

1. Choose one object and list its external interactions.
2. Draw force arrows in physically correct directions.
3. Remove velocity and net force as extra interaction arrows.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “normal force” and “tension” to explain your answer. Sentence starters are optional.

normal force: a support force perpendicular to a contact surface

tension: the pulling force through a stretched rope or cable

weight: the gravitational force on an object

01 Warm-up: bring back an earlier idea

An a-versus-F_{net} graph has slope **1.5** $(m/s^2)/N$.
1. **Find mass** and acceleration at **2.5 N**.
2. **Explain why a driving-force plot** can have an offset.

Word help: mass = a measure of how strongly an object resists acceleration

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Sigma F_x = ma_x$; $\Sigma F_y = ma_y$; $F_g = mg$.
Draw only forces on the chosen object. x and y name directions. Use g = **9.8 N/kg** unless another value is given.

Worked reasoning: A lamp hangs motionless from one vertical cable. With weight **12 N** down, the cable tension is **12 N** up. Only forces acting on the lamp belong in its diagram.

Your turn: 1. **Draw the lamp as a** labeled point with its two force arrows.
2. **Explain why its force** on the cable is not another arrow on this diagram.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



Add and label force arrows.

Lamp: draw the two 12 N forces acting on it.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Motionless lamp

A **2 kg** lamp hangs at rest from one vertical rope.
1. Use g = **9.8 N/kg**.
2. Draw and **label the forces** on the lamp, including their magnitudes.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: In my work, I used … to show … . I labeled it … .

Add and label force arrows.

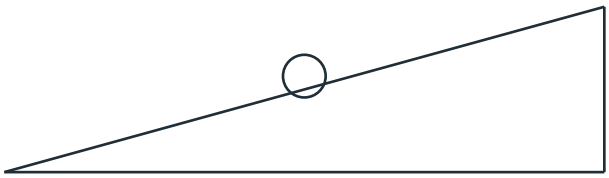
The dot represents the object. Object: 2 kg lamp

Marzano check - circle: 1 2 3

04 A rough ramp

A block rests on a rough incline.
1. **Add weight**, normal force and static friction to the ramp diagram.
2. **Explain why weight points vertically** downward rather than perpendicular to the surface.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: 1 2 3

05 Critique the extra arrows

A student's free-body diagram includes weight, normal force, friction, a "velocity force," and an extra "net force." Redraw a correct diagram for a box sliding right on a rough level floor after release. **Explain each removal**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Add and label force arrows.

Box after release; it slides right.

Marzano check - circle: 1 2 3

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Continue down each column.

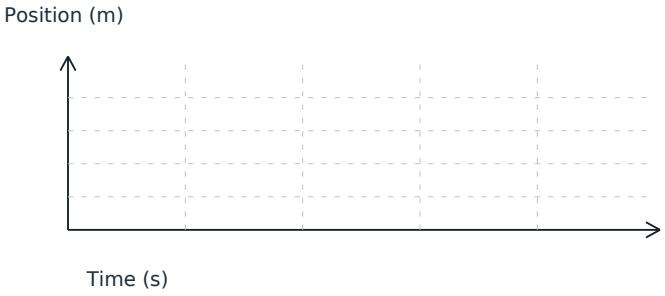
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A straight position-time segment goes from (**2 s**, **-4 m**) to (**6 s**, **20 m**). **Find velocity**, interpret its units and **predict x** at **8 s** if motion continues.
Label the diagram with the given positions and any prediction.

Word help: velocity = rate of position change, with direction

Sentence starter: I predict … . I used … to make this prediction because … .



Marzano check - circle: ① ② ③

07 Exit check

A box slides to the left on a rough level floor after release. **Draw all forces**; **state what can be concluded** about vertical and horizontal acceleration.

Sentence starter: In my work, I used … to show … . I labeled it … .



Add and label force arrows.

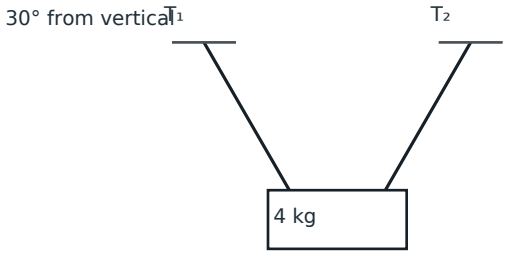
Box slides left after release on a rough level floor.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Two identical cables support a stationary **4 kg** object. Each cable is **30°** from vertical.
1. Use g = **10 N/kg**.
2. **Add force arrows** to the diagram and **calculate the tension** in each cable.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③